

An Explicit Sigma-Power Ledger for Prefix and Observability Bridges

Max-Plus Composition and Two Independence Barriers

Liang Wang

July 2026

Abstract

RH-101 removes ambient Gram assembly from the recursive packet route and RH-102 supplies an exact stopped endpoint budget. The remaining inherited gate was described only as a “prefix/normalization/observability bridge.” We replace that phrase by an exact sigma-power ledger and determine what the normalized packet mechanics can and cannot imply.

For one directional Hardy triple, write the source as $X_\sigma = N_\sigma \bar{X}_\sigma$, split at a block horizon M_σ , let P_σ be the unit-source finite-prefix norm, Z_σ the reduced packet future, R_σ the postblock packet residual, Ω_σ the future observability factor, and U_σ an upstream Hardy bridge. Exact prefix/future decomposition and the RH-77 observability theorem give

$$\mathcal{E}_\sigma \leq U_\sigma + N_\sigma (P_\sigma + Z_\sigma + \Omega_\sigma R_\sigma).$$

If a signed power $a(f)$ means $f = O(\sigma^{-a(f)} \text{polylog}(1/\sigma))$, then the directional Hardy growth power is

$$\alpha_s = \max\{0, u_s, n_s + p_s, n_s + z_s, n_s + o_s + r_s\}.$$

The two directions add: $\delta = \alpha_L + \alpha_R$. The inherited RH-49 full strict-mesh range is available when $\delta \leq 1/4$, while on $n = \sigma^{-2}L(\sigma)$ the displayed RH-54 envelope has decay power $3/4 - \delta$.

This ledger closes several bookkeeping questions. Coupling normalization, logarithmic packet rank, depth-five memory, the fixed stopped gate, and mesh substitution all carry zero sigma power. Observation growth may be cancelled by residual decay because $o_s + r_s$ is a signed sum; terms within one direction combine by a maximum, not by addition.

It also gives a negative theorem. A nilpotent two-dimensional family has normalized packet Gram equal to one and zero postblock tail, yet finite-prefix Hardy power equal to any prescribed $a > 0$. A scalar stable family likewise has a perfect rank-one packet at every time while observation scaling creates any prescribed Hardy power $b > 0$. Thus a uniform quotient law, structured Gram action, and stopped clock cannot by themselves prove the prefix or absolute observability estimates.

The five-anchor ledger remains fully green: conditional Hardy uppers are at most 1.836, rank clocks are at most seven, depth five and the stopped 1.01 gate persist, and the stress identification envelope decreases from 0.07354 to 0.002961. These finite facts do not close the new all-level scalar leaves. Hence the optimistic claim that only the quotient gate remains must be revised. No unconditional Stage A, moving-cloud construction, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: sigma-power ledger; max-plus algebra; Hardy energy; observability Gramian; normalized packet; finite-prefix transient.

MSC 2020: 47A55; 93B28; 65F15; 37M25; 15A18.

1 Introduction

The packet corridor now has unusually explicit finite-dimensional mechanics. Source-seeded refresh avoids late ambient eigenspaces, reduced moments avoid ambient cross factorizations,

finite-memory state actions avoid ambient Gramians, and an exact stopped clock prevents locally certified quotients from overspending the endpoint gate [3, 6, 7]. These are genuine advances, but they concern relative packet geometry.

Stage A ultimately needs an absolute directional Hardy norm. The RH-100 route review retained a separate gate O for finite prefix, normalization, and observability, then proposed RH-103 to expose its powers rather than hide them under “polylogarithmic” [5]. That distinction is essential: normalization can erase a scalar amplitude from every packet Gramian, while the same amplitude still appears in an unnormalized transfer sequence or observation.

This paper has two goals. First, it proves the exact max-plus sigma-power formula for the effective-rank corridor. Second, it tests whether the packet gates can imply the missing terms. Two elementary stable families show that they cannot. The result is not a failure of the packet route; it is a sharper dependency graph. The remaining wall consists of explicit scalar exponent laws that can be attacked separately.

2 Directional prefix/future decomposition

Let $A_\sigma \in \mathbb{C}^{d \times d}$, source X_σ , and observation Y_σ . The directional Hardy norm is

$$\mathcal{E}_\sigma^2 = \sum_{j \geq 0} \|Y_\sigma A_\sigma^j X_\sigma\|_F^2. \quad (1)$$

Write

$$X_\sigma = N_\sigma \bar{X}_\sigma, \quad \|\bar{X}_\sigma\|_F = 1,$$

where N_σ includes the actual normalized coupling/range factor used by the physical triple. At a block horizon $M = M_\sigma$, define

$$P_\sigma^2 = \sum_{j=0}^{M-1} \|Y_\sigma A_\sigma^j \bar{X}_\sigma\|_F^2, \quad (2)$$

$$\bar{B}_\sigma = A_\sigma^M \bar{X}_\sigma. \quad (3)$$

Let $\bar{B}_{\sigma,r}$ be a rank- r approximation and put

$$Z_\sigma = T_\sigma(\bar{B}_{\sigma,r}), \quad (4)$$

$$R_\sigma = \|\bar{B}_\sigma - \bar{B}_{\sigma,r}\|_F. \quad (5)$$

Here T_σ is the complete future Hardy seminorm. If $q_\sigma = \|A_\sigma^M\|_2 < 1$, the RH-77 block-observability theorem gives

$$\Omega_\sigma = \sqrt{\frac{\|O_{\sigma,M}\|_2}{1 - q_\sigma^2}}, \quad |T_\sigma(\bar{B}_\sigma) - Z_\sigma| \leq \Omega_\sigma R_\sigma. \quad (6)$$

Finally, let U_σ be a Hardy-norm upper for the analytic/frozen or upstream/reference transfer difference. RH-74 constructs this object at the five anchors [8].

Theorem 2.1 (Explicit prefix–packet–observability ledger). *The complete directional norm satisfies*

$$\boxed{\mathcal{E}_\sigma \leq U_\sigma + N_\sigma(P_\sigma + Z_\sigma + \Omega_\sigma R_\sigma)}. \quad (7)$$

Proof. The triangle inequality separates the upstream/reference bridge. For the reference triple, split (1) before and after M . The finite part is $N_\sigma P_\sigma$. Apply the reverse triangle inequality for the future seminorm and (6) to obtain $N_\sigma(Z_\sigma + \Omega_\sigma R_\sigma)$. $\square$

The theorem deliberately keeps the observability factor and residual separate. One may grow while the other decays. Multiplying two separately fitted absolute powers only after their signs are recorded prevents a common bookkeeping error.

3 Signed sigma powers and max-plus composition

Definition 3.1 (Signed sigma power). For a nonnegative family f_σ , a signed ledger power $a_f \in \mathbb{R}$ means

$$f_\sigma = O(\sigma^{-a_f} \text{polylog}(1/\sigma)).$$

A negative a_f records decay. The final Hardy growth exponent is truncated below at zero.

Let $u_s, n_s, p_s, z_s, o_s, r_s$ be the powers of the six terms in (7) for side $s \in \{L, R\}$.

Theorem 3.2 (Max-plus directional Hardy power theorem). *The directional growth power may be chosen as*

$$\alpha_s = \max\{0, u_s, n_s + p_s, n_s + z_s, n_s + o_s + r_s\}. \quad (8)$$

For the RH-54 two-sided composition,

$$\delta = \alpha_L + \alpha_R, \quad \|\mathcal{I}_{n,\sigma}\|_{S_2} = O(n^{-2} \sigma^{-13/4-\delta} \text{polylog}(1/\sigma)). \quad (9)$$

The inherited RH-49 full strict-mesh range is green when

$$\delta \leq \frac{1}{4}. \quad (10)$$

On the displayed stress schedule $n = \sigma^{-2}L(\sigma)$, the envelope in (9) has sigma decay power

$$\frac{3}{4} - \delta. \quad (11)$$

Proof. Products add signed powers and finite sums take their maximum. Apply those two rules to (7), including zero because a bounded Hardy norm has growth power zero. RH-54 proves that the two directional powers add in the Hilbert–Schmidt gain and yields (9) [1]. The RH-49 stable-rank threshold is (10). Substituting $n = \sigma^{-2}L(\sigma)$ gives (11). $\square$

Remark 3.3. Terms within one side take a maximum, not a sum. Prefix growth and reduced future growth are alternative contributions to the same norm. By contrast, the left and right Hardy powers add because the downstream directional gain contains their product.

Remark 3.4. The stress exponent can remain positive when $\delta > 1/4$. That does not override the established RH-49 claim boundary: the quarter-power condition is the sufficient threshold for retaining the full strict mesh range, not merely one chosen stress schedule.

4 Zero-power overheads already closed

Several quantities in the packet architecture do not consume the quarter budget.

1. The couplings B and C are normalized in Hilbert–Schmidt norm. RH-52 bounds the required range-restricted residues, and RH-54 proves that normalization converts the adaptive cutoff defect into a decaying relative error. Thus $n_s = 0$ in the intended ledger [1, 2].
2. The half-logarithmic rank $r_\sigma = O(\log(1/\sigma))$ carries zero sigma power [4].
3. The RH-101 common memory depth is five and $\eta = 1/512$, both constants.
4. The RH-102 target 1.01 and safety fraction 0.99 are constants.
5. Replacing n by $\sigma^{-2}L(\sigma)$ is exact algebra; logarithmic L does not create a sigma power.

These facts are useful, but they do not bound P_σ , Z_σ , or $\Omega_\sigma R_\sigma$.

Table 1: Finite-anchor quantities. The observed residual product already includes future observability; its small size does not bound either factor uniformly.

σ	prefix	$\ O\ ^{1/2}$	observed residual	bridge	rank	Hardy
0.16	1.003	4.097	$5.04 \cdot 10^{-9}$	$1.42 \cdot 10^{-6}$	4	1.078
0.08	1.265	6.076	$2.11 \cdot 10^{-12}$	$8.13 \cdot 10^{-7}$	5	1.309
0.04	1.485	9.299	$7.26 \cdot 10^{-13}$	$2.02 \cdot 10^{-6}$	6	1.506
0.02	1.634	14.099	$6.26 \cdot 10^{-12}$	$6.63 \cdot 10^{-7}$	6	1.679
0.01	1.760	21.255	$1.75 \cdot 10^{-12}$	$3.12 \cdot 10^{-7}$	7	1.836

5 Two independence barriers

The next propositions show that the missing terms cannot be recovered from perfect normalized packets alone.

Proposition 5.1 (Finite-prefix independence barrier). *For every $a > 0$, let*

$$A_\sigma = \begin{pmatrix} 0 & \sigma^{-a} \\ 0 & 0 \end{pmatrix}, \quad X = e_2, \text{quad} Y = e_1^*.$$

Then $A_\sigma^2 = 0$ and

$$\mathcal{E}_\sigma = \sigma^{-a}. \quad (12)$$

Every nonzero state has one source coordinate, so its normalized packet Gramian is the scalar 1 and its rank-one relative tail is zero. The postblock future after two steps is also zero.

Proof. $YX = 0$, $YA_\sigma X = \sigma^{-a}$, and all later responses vanish. This gives (12). A nonzero one-column state S satisfies $S^*S / \|S\|_F^2 = [1]$, so normalized packet data contain no trace of the transient amplitude. $\square$

Proposition 5.2 (Observation independence barrier). *For every $b > 0$, take the scalar stable system*

$$A = c, \text{quad} X = 1, \text{quad} Y_\sigma = \sigma^{-b}, \text{quad} 0 < c < 1.$$

Every normalized state Gramian is 1 and every rank-one packet tail is zero, while

$$\mathcal{E}_\sigma = \frac{\sigma^{-b}}{\sqrt{1 - c^2}}. \quad (13)$$

Proof. The packet statement is immediate in one dimension. Summing the scalar geometric Hardy series gives (13). $\square$

Both systems are stable; the first is nilpotent. The propositions do not say that the physical folded-Gaussian family has these bad powers. They prove a logical separation: Q , G , and H_{stop} cannot imply the missing prefix and observation laws without additional physical input.

6 Five-anchor ledger

The machine-readable audit combines RH-74, RH-75, RH-77, RH-78, RH-82, RH-101, and RH-102 without refitting an all-level law. Selected quantities are shown in Table 1.

The absolute observability factor grows from 4.10 to 21.26, while the observation-weighted clock residual remains below 5.05×10^{-9} . This is precisely why the signed sum $o_s + r_s$ is the correct ledger entry. A separate theorem must show that the residual decay compensates observation growth at every level.

The depth-five action passes, the primary stopped endpoint ratio is at most 1.00117243, and every finite anchor remains inside the conditional zero-power Hardy envelope. On the RH-78 stress mesh, the archived identification envelope decreases from 0.07354 to 0.002961.

Table 2: Illustrative exact max-plus scenarios. These are power allocations, not fits to the physical family.

scenario	α_L	α_R	δ	quarter gate
zero-power target	0	0	0	green
balanced boundary	0.125	0.125	0.25	green
one-sided observation leak	0.30	0	0.30	red
two-sided prefix leak	0.16	0.12	0.28	red
observation/residual cancellation	0	0	0	green
max-not-sum example	0.12	0.10	0.22	green

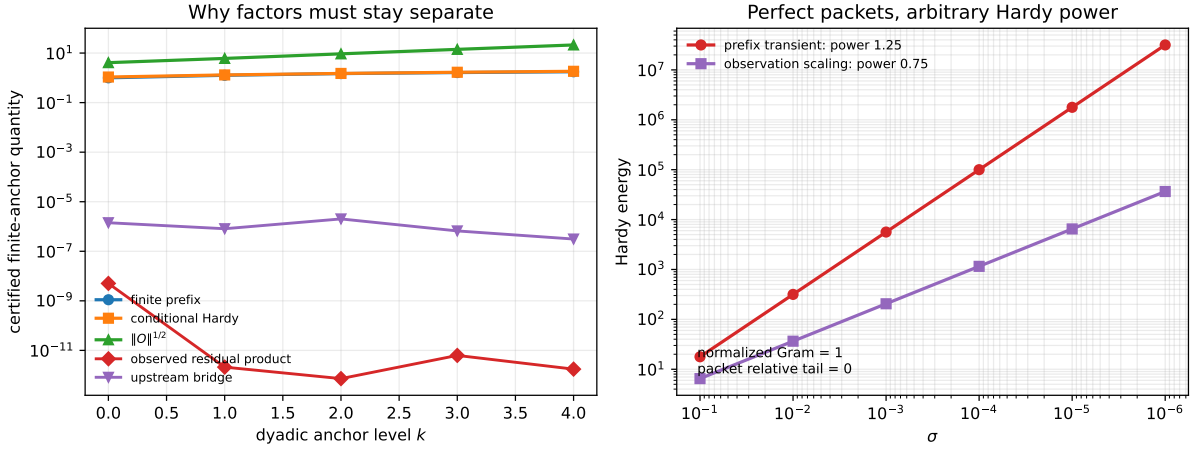


Figure 1: Left: finite factors have very different scales and must remain separate in the asymptotic ledger. Right: two exact stable families keep normalized packet tails zero while assigning arbitrary Hardy powers.

7 Revised Stage-A frontier

The RH-100 optimistic branch said that positive RH-101–RH-103 results could leave only the uniform quotient law Q . RH-101 and RH-102 are positive in their stated scopes. RH-103 is mixed:

- normalization, rank, memory, stopping, and mesh bookkeeping are closed at zero sigma power;
- the exact exponent formula is closed;
- the five-anchor composition is green;
- the all-level prefix and absolute observation-weighted future laws are logically independent of normalized packet success.

Therefore O must not be deleted. It should be replaced by named scalar leaves corresponding to u_s , p_s , z_s , and $o_s + r_s$. Some future formulation of Q may also provide part of r_s ; it still cannot provide p_s or o_s by itself, as Propositions 5.1 and 5.2 show.

The fallback RH-75 full-block law remains valid: its square-root contraction explicitly supplies the observation/residual cancellation together with a finite-prefix companion. The packet route remains attractive because it can attack weaker directional terms, but it now has a more honest list of them.

No unconditional Stage A follows from this ledger. Moving-cloud Riesz projections, coefficient bridges, trace-class complements, self-adjoint counting, arithmetic trace formulas, and zero identification remain downstream. In particular, no Hilbert–Polya operator or proof of the Riemann Hypothesis is claimed.

8 Conclusion

The word “polylogarithmic” has been replaced by an exact max-plus formula. Within one directional Hardy norm, alternative contributions take a maximum; observation and residual powers add with sign; the two directions then add against one shared quarter-power budget. Several packet overheads cost no sigma power at all.

The same analysis prevents an overclaim. Perfect normalized packets do not bound finite transients or absolute observation scale. The next subprogram should therefore prove a physical finite-prefix law and an observation–residual cancellation law, while continuing the uniform gap-aware quotient attack. That is a narrower and more testable frontier than the former single letter O .

References

- [1] Liang Wang. Factor-aware sparse-to-full transfer for intrinsic riesz identification. 2026. RH-54 manuscript.
- [2] Liang Wang. Intrinsic peripheral residue transfer from weak finite factors. 2026. RH-52 manuscript.
- [3] Liang Wang. Finite-memory packet actions for normalized gram recursions. 2026. RH-101 manuscript.
- [4] Liang Wang. Exponential excess-rank tails and the half-logarithmic postblock clock. 2026. RH-82 manuscript.
- [5] Liang Wang. A hundred-layer audit of prime dynamics. 2026. RH-100 manuscript.
- [6] Liang Wang. Source-seeded four-direction horizon refresh. 2026. RH-94 manuscript.
- [7] Liang Wang. Stopped hybrid quotient clocks. 2026. RH-102 manuscript.
- [8] Liang Wang. Validated upstream-to-frozen hardy bridges. 2026. RH-74 manuscript.